Q1. Who is the senior-most employee based on job title?

```
SELECT * FROM employee
ORDER BY levels DESC
LIMIT 1;
```

OUTPUT:

Data Output Messages Notifications

Showing rows: 1 to 1 Page No: 1 of 1

	employee_id [PK] character varying (50)	last_name character (50)	first_name character (50)	title character varying (50)	reports_to character varying (30)	levels character varying (10)	birthdate timestamp without time zone	hire_date timestamp w
1	9	Madan	Mohan	Senior General Manag...	[null]	L7	1961-01-26 00:00:00	2016-01-14

Q2. Which countries have the most Invoices?

```
SELECT COUNT(*) AS C, billing_country
FROM invoice
GROUP BY billing_country
ORDER BY C DESC;
```

OUTPUT:

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	c bigint	billing_country character varying (30)
1	131	USA
2	76	Canada
3	61	Brazil
4	50	France
5	41	Germany
6	30	Czech Republic
7	29	Portugal
8	28	United Kingdom
9	21	India
10	13	Chile
11	13	Ireland
12	11	Spain
13	11	Finland
14	10	Australia
15	10	Netherlands
16	10	Sweden
17	10	Poland
18	10	Hungary
19	10	Denmark
20	9	Austria
21	9	Norway

21	9	Norway
22	9	Italy
23	7	Belgium
24	5	Argentina

Q3. What are top 3 values of total invoice?

```
SELECT total FROM invoice
ORDER BY total DESC
LIMIT 3;
```

OUTPUT:

Data Output Messages Notifications

	total double precision
1	23.759999999999998
2	19.8
3	19.8

Q4. Find one city that has the highest sum of invoice totals. Return both the city name & sum of all invoice totals.

```
SELECT SUM(total) AS invoice_total, billing_city
FROM invoice
GROUP BY billing_city
ORDER BY invoice_total DESC;
```

OUTPUT:

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	invoice_total double precision	billing_city character varying (30)
1	273.24000000000007	Prague
2	169.29	Mountain View
3	166.32	London
4	158.4	Berlin
5	151.47	Paris
6	129.69	São Paulo
7	114.83999999999997	Dublin
8	111.86999999999999	Delhi
9	108.89999999999998	São José dos Campos
10	106.91999999999999	Brasília
11	102.96000000000001	Lisbon
12	99.99	Bordeaux
13	99.99	Montréal
14	98.01	Madrid
15	98.01	Redmond
16	97.02000000000001	Santiago
17	94.05000000000001	Frankfurt
18	92.07000000000001	Orlando
19	91.08000000000001	Reno
20	91.08	Ottawa
21	86.13000000000002	Fort Worth

Q5. Find the person who has spent the most money.

```
SELECT c.customer_id, c.first_name, c.last_name, SUM(i.Total) AS Total_amount FROM
customer c
INNER JOIN invoice i ON c.customer_id = i.customer_id
GROUP BY c.customer_id
ORDER BY Total_amount DESC
LIMIT 1;
```

OUTPUT:

Data Output Messages Notifications

	customer_id [PK] integer	first_name character (50)	last_name character (50)	total_amount double precision
1	5	R ...	Madhav ...	144.54000000000002

Q6. Write a query to return the email, first name, last name, & Genre of all Rock Music listeners. Return your list ordered alphabetically by email, starting with A.

```
SELECT DISTINCT email, first_name, last_name
FROM customer c
JOIN invoice i on c.customer_id = i.customer_id
JOIN invoice_line il on i.invoice_id = il.invoice_id
WHERE track_id IN (
   SELECT track_id FROM track
         JOIN genre ON track.genre_id = genre.genre_id
         WHERE genre.name = 'Rock'
)
ORDER BY email;
```

OUTPUT:

Data Output Messages Notifications

	email character varying (50)	first_name character (50)	last_name character (50)
1	aaronmitchell@yahoo.ca	Aaron	Mitchell
2	alero@uol.com.br	Alexandre	Rocha
3	astrid.gruber@apple.at	Astrid	Gruber
4	bjorn.hansen@yahoo.no	Bjørn	Hansen
5	camille.bernard@yahoo.fr	Camille	Bernard
6	daan_peeters@apple.be	Daan	Peeters
7	diego.gutierrez@yahoo.ar	Diego	Gutiérrez
8	dmiller@comcast.com	Dan	Miller
9	dominiquelefebvre@gmail....	Dominique	Lefebvre
10	edfrancis@yachoo.ca	Edward	Francis
11	eduardo@woodstock.com.br	Eduardo	Martins
12	ellie.sullivan@shaw.ca	Ellie	Sullivan
13	emma_jones@hotmail.com	Emma	Jones
14	enrique_munoz@yahoo.es	Enrique	Muñoz
15	fernadaramos4@uol.com.br	Fernanda	Ramos
16	fharris@google.com	Frank	Harris
17	fralston@gmail.com	Frank	Ralston
18	ftremblay@gmail.com	François	Tremblay
19	fzimmermann@yahoo.de	Fynn	Zimmermann
20	hannah.schneider@yahoo.de	Hannah	Schneider
21	hholy@gmail.com	Helena	Holý

Total rows: 59 Query complete 00:00:00.274

Q7. Let's invite the artists who have written the most rock music in our dataset. Find the artist's name and total track count of the top 10 rock bands.

```
SELECT ar.artist_id, ar.name, COUNT(g.name) AS Number_of_Songs
FROM artist ar
JOIN album al ON ar.artist_id = al.artist_id
JOIN track t ON al.album_id = t.album_id
JOIN genre g ON t.genre_id = g.genre_id
```

```
WHERE g.name = 'Rock'
GROUP BY ar.artist_id
ORDER BY Number_of_Songs DESC
LIMIT 10;
```

OUTPUT:

Data Output Messages Notifications

	artist_id [PK] character varying (50)	name character varying (120)	number_of_songs bigint
1	22	Led Zeppelin	114
2	150	U2	112
3	58	Deep Purple	92
4	90	Iron Maiden	81
5	118	Pearl Jam	54
6	152	Van Halen	52
7	51	Queen	45
8	142	The Rolling Stones	41
9	76	Creedence Clearwater Revi...	40
10	52	Kiss	35

Q8 Return all the track names that have a song length longer than the average song length. Return the name, and Milliseconds for each track. Order by the song length with the longest songs listed first.

```
SELECT name, milliseconds
FROM track
WHERE milliseconds > (SELECT AVG(milliseconds) AS Avg_lenght FROM track)
ORDER BY milliseconds DESC;
SELECT * FROM track;
```

OUTPUT:

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	name character varying (150)	milliseconds integer
1	Occupation / Precipice	5286953
2	Through a Looking Glass	5088838
3	Greetings from Earth, Pt. 1	2960293
4	The Man With Nine Lives	2956998
5	Battlestar Galactica, Pt. 2	2956081
6	Battlestar Galactica, Pt. 1	2952702
7	Murder On the Rising Star	2935894
8	Battlestar Galactica, Pt. 3	2927802
9	Take the Celestra	2927677
10	Fire In Space	2926593
11	The Long Patrol	2925008
12	The Magnificent Warriors	2924716
13	The Living Legend, Pt. 1	2924507
14	The Gun On Ice Planet Zero, Pt. 2	2924341
15	The Hand of God	2924007
16	Experiment In Terra	2923548
17	War of the Gods, Pt. 2	2923381
18	The Living Legend, Pt. 2	2923298
19	War of the Gods, Pt. 1	2922630
20	Lost Planet of the Gods, Pt. 1	2922547
21	Baltar's Escape	2922088

Total rows: 494 Query complete 00:00:00.236

Q9 Find how much amount was spent by each customer on artists? Write a query to return the customer name, artist name, and total spent.

```
WITH best_selling_artist AS(
   SELECT artist.artist_id AS artist_id, artist.name AS artist_name,
         SUM(invoice_line.unit_price * invoice_line.quantity) AS total_sales
         FROM invoice_line
         JOIN track ON track.track_id = invoice_line.track_id
         JOIN album ON album.album_id = track.album_id
         JOIN artist ON artist.artist_id = album.artist_id
```

```
        GROUP BY 1
        ORDER BY 3 DESC
        LIMIT 1
)
-- SELECT * FROM best_selling_artist;
SELECT c.customer_id, c.first_name, c.last_name,bsa.artist_name,
SUM(il.unit_price*il.quantity) AS amount_spent
FROM invoice i
JOIN customer c ON c.customer_id = i.customer_id
JOIN invoice_line il ON il.invoice_id = i.invoice_id
JOIN track ON track.track_id = il.track_id
JOIN album ON album.album_id = track.album_id
JOIN best_selling_artist bsa ON bsa.artist_id = bsa.artist_id
GROUP BY 1,2,3,4
ORDER by 5 DESC;
```

OUTPUT:

Data Output Messages Notifications

	customer_id integer	first_name character (50)	last_name character (50)	artist_name character varying (120)	amount_spent double precision
1	5	R	Madhav	Queen	144.53999999999985
2	6	Helena	Holý	Queen	128.6999999999997
3	46	Hugh	O'Reilly	Queen	114.83999999999978
4	58	Manoj	Pareek	Queen	111.86999999999979
5	1	Luís	Gonçalves	Queen	108.8999999999998
6	13	Fernanda	Ramos	Queen	106.91999999999982
7	34	João	Fernandes	Queen	102.95999999999984
8	3	François	Tremblay	Queen	99.98999999999985
9	42	Wyatt	Girard	Queen	99.98999999999985
10	50	Enrique	Muñoz	Queen	98.00999999999986
11	53	Phil	Hughes	Queen	98.00999999999986
12	17	Jack	Smith	Queen	98.00999999999986
13	57	Luis	Rojas	Queen	97.01999999999987
14	20	Dan	Miller	Queen	95.03999999999988
15	37	Fynn	Zimmermann	Queen	94.04999999999988
16	22	Heather	Leacock	Queen	92.0699999999999
17	30	Edward	Francis	Queen	91.0799999999999
18	21	Kathy	Chase	Queen	91.0799999999999
19	26	Richard	Cunningham	Queen	86.12999999999992
20	36	Hannah	Schneider	Queen	85.13999999999993
21	27	Patrick	Gray	Queen	84.14999999999993

Total rows: 59 Query complete 00:00:00.291

Q10. We want to find out the most popular music Genre for each country. We determine the most popular genre as the genre with the highest amount of purchases. Write a query that returns each country along with the top Genre. For countries where the maximum number of purchases is shared, return all genres.

```
WITH cte_popular_genre AS (
    SELECT COUNT(il.*) AS purchases, c.country, g.name, g.genre_id,
        ROW_NUMBER() OVER(PARTITION BY c.country ORDER BY COUNT(il.quantity)
DESC) AS RowNo
        FROM invoice_line il
        JOIN invoice i ON il.invoice_id = i.invoice_id
        JOIN customer c ON i.customer_id = c.customer_id
        JOIN track t ON t.track_id = il.track_id
        JOIN genre g ON g.genre_id = t.genre_id
        GROUP BY 2, 3, 4
        ORDER BY 2 ASC, 1 DESC
)
SELECT *
FROM cte_popular_genre
WHERE RowNo <= 1
ORDER BY purchases DESC
LIMIT 10;
```

OUTPUT:

Data Output Messages Notifications

	purchases bigint	country character varying (50)	name character varying (120)	genre_id character varying (50)	rowno bigint
1	561	USA	Rock	1	1
2	333	Canada	Rock	1	1
3	211	France	Rock	1	1
4	205	Brazil	Rock	1	1
5	194	Germany	Rock	1	1
6	166	United Kingdom	Rock	1	1
7	143	Czech Republic	Rock	1	1
8	108	Portugal	Rock	1	1
9	102	India	Rock	1	1
10	72	Ireland	Rock	1	1

Q11. Write a query that determines the customer who has spent the most on music for each country. Write a query that returns the country along with the top customer, and how much they spent. For countries where the top amount spent is shared, provide all customers who spent this amount.

```
WITH CTE_spending_byCustomer AS(
    SELECT c.customer_id, c.first_name, c.last_name, i.billing_country, SUM(i.total) AS
total_spending,
    ROW_NUMBER() OVER(PARTITION BY billing_country ORDER BY SUM(i.total) DESC)
AS RowNo
    FROM customer c
    JOIN invoice i ON i.customer_id = c.customer_id
    GROUP BY 1, 2, 3,4
```

```
    ORDER BY 4 ASC, 5 DESC
)
SELECT * FROM CTE_spending_byCustomer
WHERE RowNo <= 1;
```

OUTPUT:

Data Output Messages Notifications

Showing rows: 1

	customer_id integer	first_name character (50)	last_name character (50)	billing_country character varying (30)	total_spending double precision	rowno bigint
1	56	Diego	Gutiérrez	Argentina	39.6	1
2	55	Mark	Taylor	Australia	81.18	1
3	7	Astrid	Gruber	Austria	69.3	1
4	8	Daan	Peeters	Belgium	60.38999999999999	1
5	1	Luís	Gonçalves	Brazil	108.89999999999998	1
6	3	François	Tremblay	Canada	99.99	1
7	57	Luis	Rojas	Chile	97.02000000000001	1
8	5	R	Madhav	Czech Republic	144.54000000000002	1
9	9	Kara	Nielsen	Denmark	37.61999999999999	1
10	44	Terhi	Hämäläinen	Finland	79.2	1
11	42	Wyatt	Girard	France	99.99	1
12	37	Fynn	Zimmermann	Germany	94.05000000000001	1
13	45	Ladislav	Kovács	Hungary	78.21	1
14	58	Manoj	Pareek	India	111.86999999999999	1
15	46	Hugh	O'Reilly	Ireland	114.83999999999997	1
16	47	Lucas	Mancini	Italy	50.49	1
17	48	Johannes	Van der Berg	Netherlands	65.34	1
18	4	Bjørn	Hansen	Norway	72.27000000000001	1
19	49	Stanisław	Wójcik	Poland	76.22999999999999	1
20	34	João	Fernandes	Portugal	102.96000000000001	1
21	50	Enrique	Muñoz	Spain	98.01	1

Total rows: 24 Query complete 00:00:00.241